



Computer Network

Dynamic Host Configuration Protocol
(DHCP)



What is DHCP's purpose?



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- DHCP's purpose is to enable individual computers on an IP network to extract their configurations from a server (the 'DHCP server') or servers, in particular, servers that have no exact information about the individual computers until they request the information.



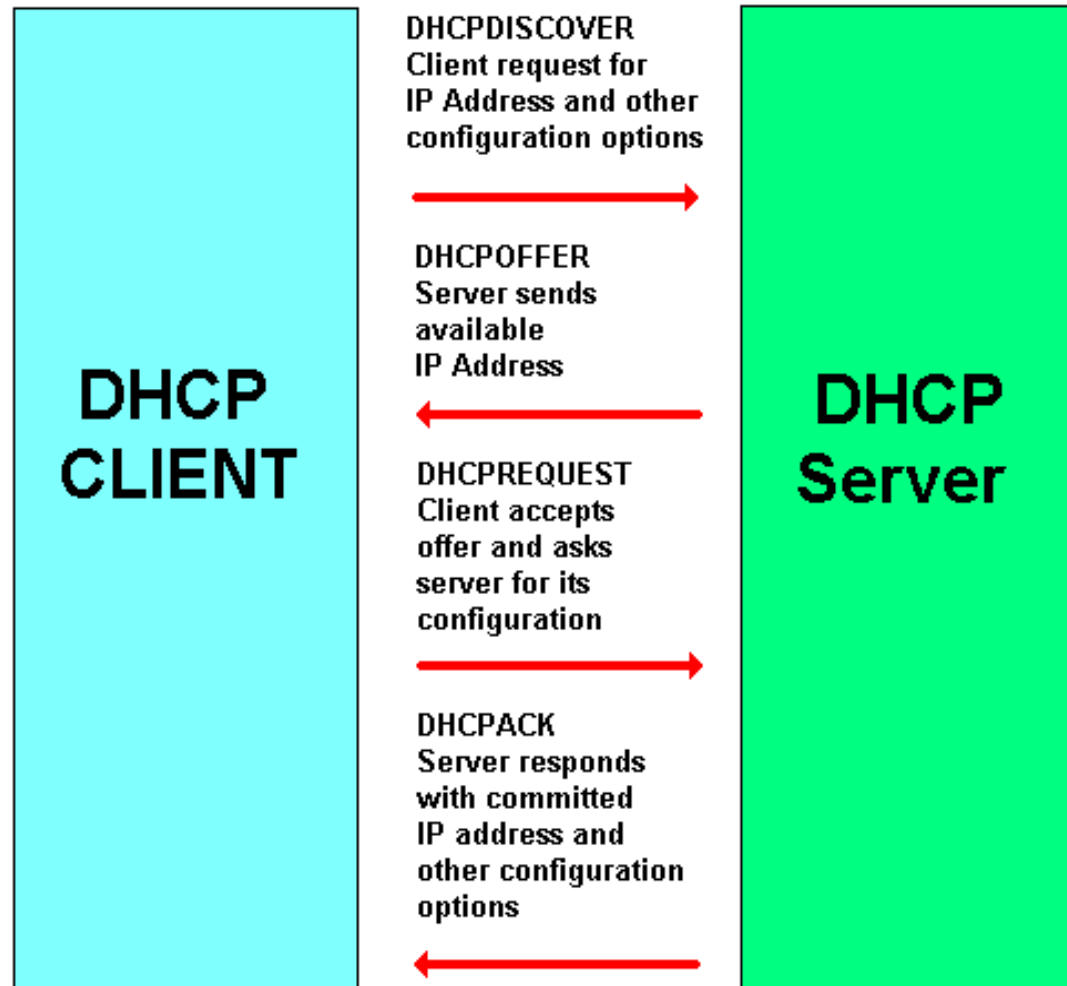
How Does DHCP Work?



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- When a client needs to start up TCP/IP operations, it broadcasts a request for address information. The DHCP server receives the request, assigns a new address for a specific time period (called a lease period) and sends it to the client together with the other required configuration information



How Does DHCP Work?





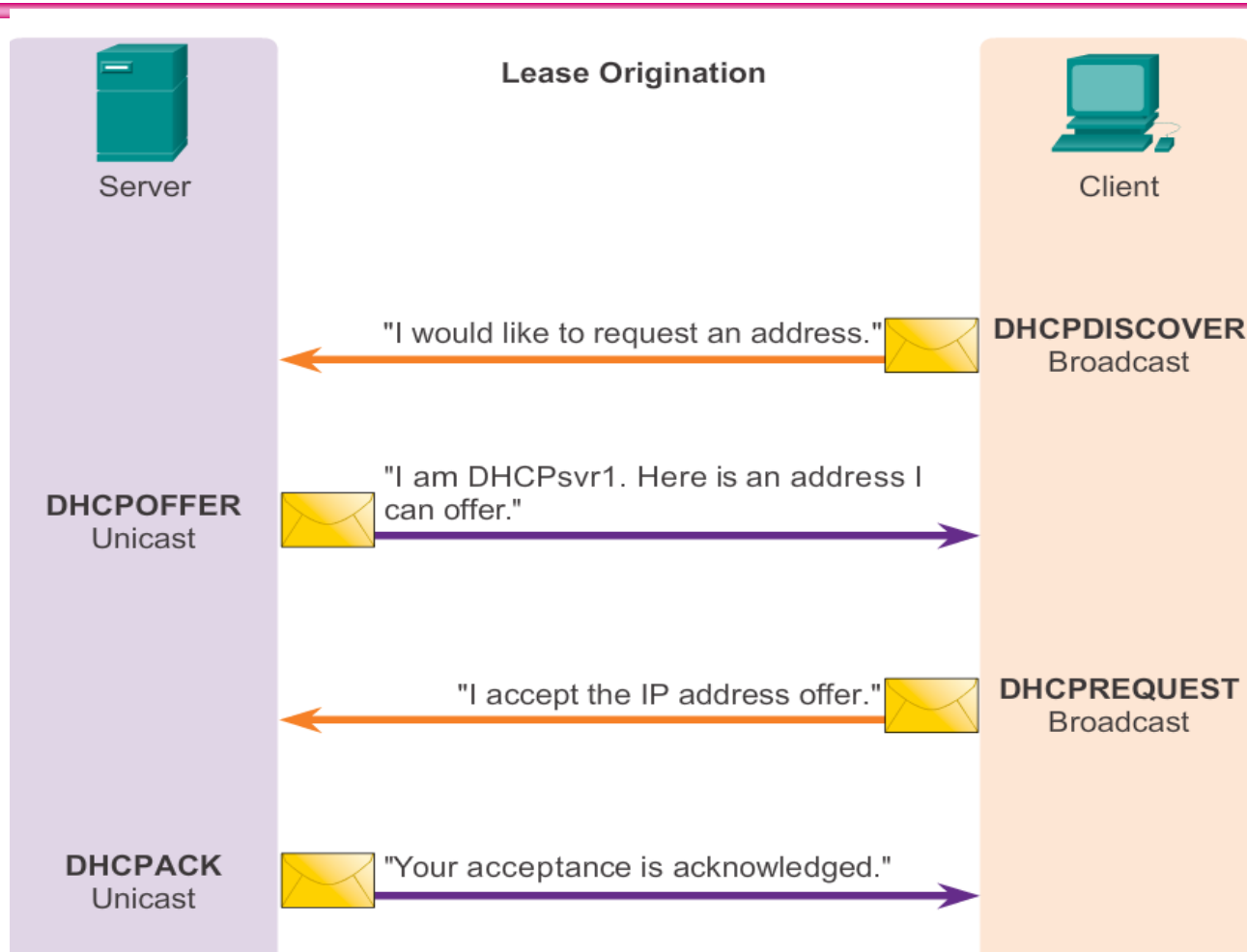
What protocol and port does DHCP use?



DHCP runs over UDP, utilizing ports 67 and 68.



How the DHCP Lease Generation Process Works





DHCP configuration

- DHCP server
 - Router(config)#**ip dhcp pool** *name*
 - Router(dhcp-config)#**network** *network mask*
 - Router(config)#**ip dhcp excluded-address** *start-ip end-ip*
- DHCP client
 - Router(config)#**interface** interface
 - Router(config-if)#**ip address dhcp**



DHCP configuration

- Router(config)#ip dhcp pool Net1
- Router(dhcp-config)#network 192.168.1.0 255.255.255.0
- Router(dhcp-config)#default-router 192.168.1.1
- Router(dhcp-config)#dns-server 203.203.203.203
- Router(config-if)#exit
- Router(config)#ip dhcp excluded-address 192.168.1.1
- Router(config)#interface fa0/0
- Router(config-if)#ip address 192.168.1.1 255.255.255.0
- Router(config-if)#no shut